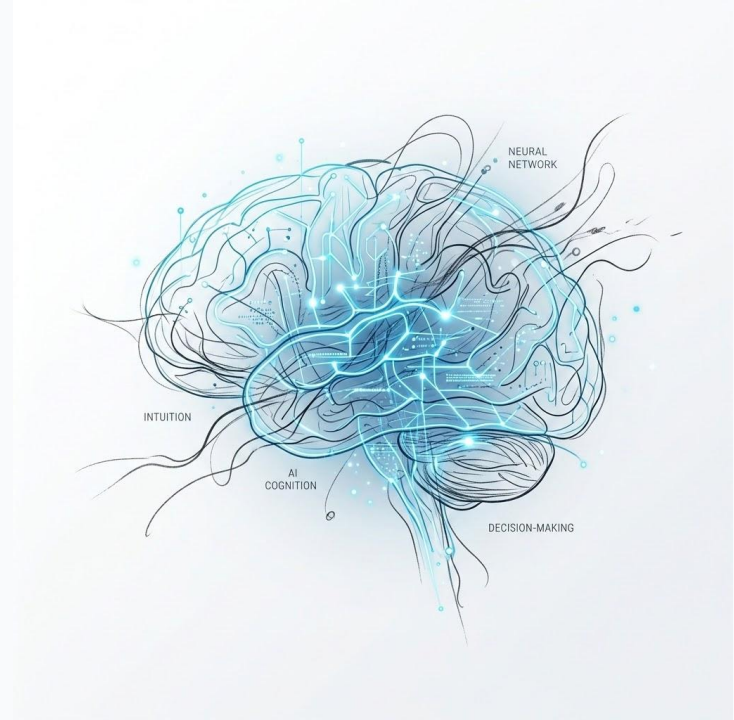


How do humans learn?

Psychological Foundations

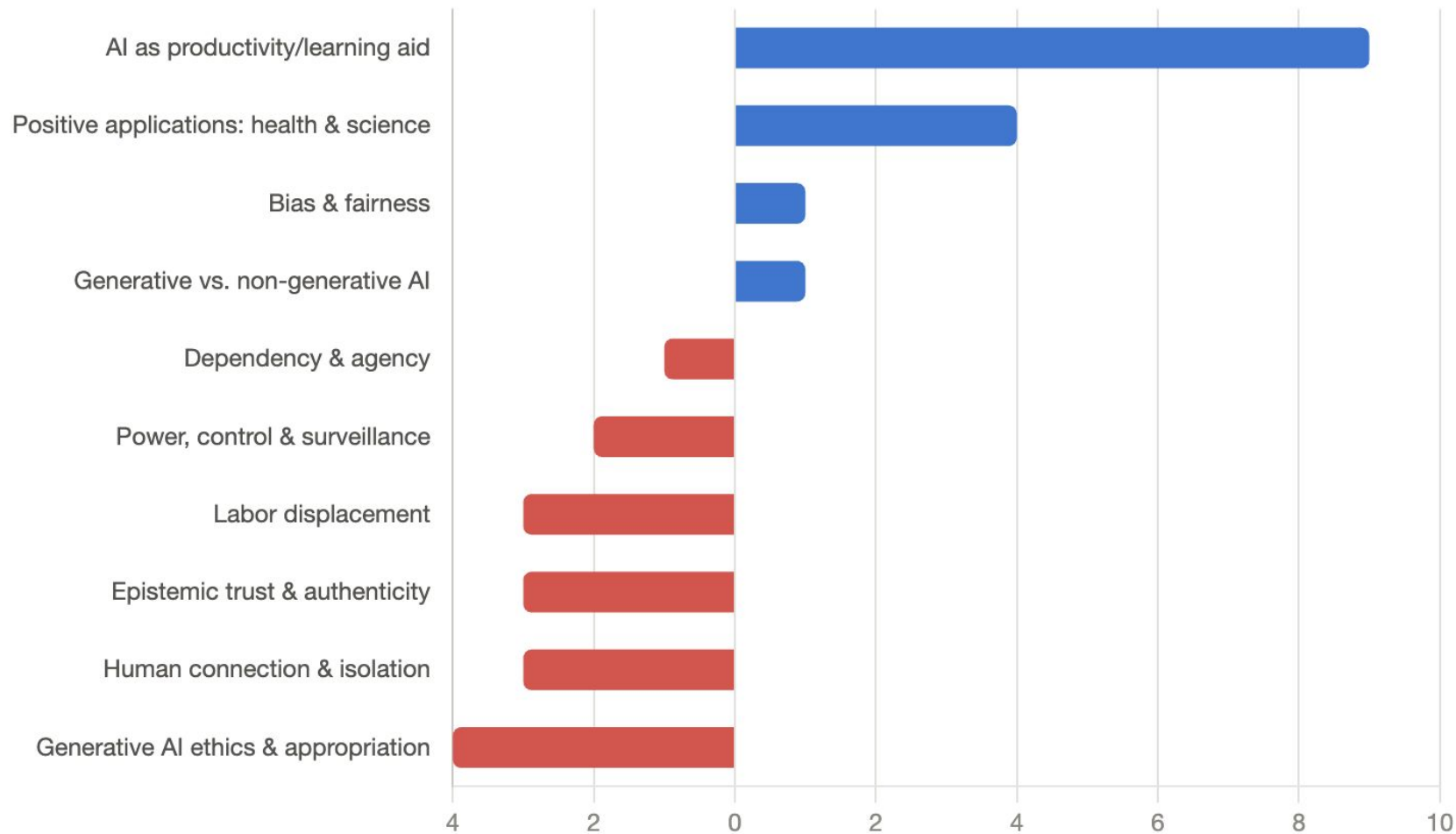
SYS 478. Fall 2026. Prof. Sarah Van Wart



Outline

1. Forum Reflection
2. Warm-up
3. Behaviorism
4. Constructivism

■ Coded as "valuable" ■ Coded as "gives pause"



One question I hope this course helps me answer

- **Ethical / responsible use:** Where is the line between helpful tool, dependence, and harm (and what contexts are appropriate / inappropriate)?
- **Systemic effects:** What are AI's broader political, environmental, economic, and social consequences?
- **Power & accountability:** Who is shaping our behavior, and who is responsible?
- **Trust:** When should we believe AI outputs?
- **The future:** Is AI's current trajectory inevitable – and what alternatives are possible?

Activity 1: Guess the Goal

Debrief

- What caused their behavior to change?
- Did they understand the goal, or just become better at producing the rewarded behavior?
- Would they know when to reproduce the behavior later?
- Your thoughts here...

Behaviorism

Behaviorists define learning as a change in observable behavior shaped by the environment.

- Focuses on observable behavior, not internal mental states
- Behaviors become more or less likely based on their consequences
- Reinforcement increases a behavior
- Punishment decreases a behavior

Implications for learning: What conditions make a behavior more likely to happen again?

Behaviorism: Example

Rewards

- A child is rewarded (extra play time) if they do their homework
- Over time, the child associates completing homework with positive outcomes, which increases the likelihood of them doing it again in the future.

Punishments

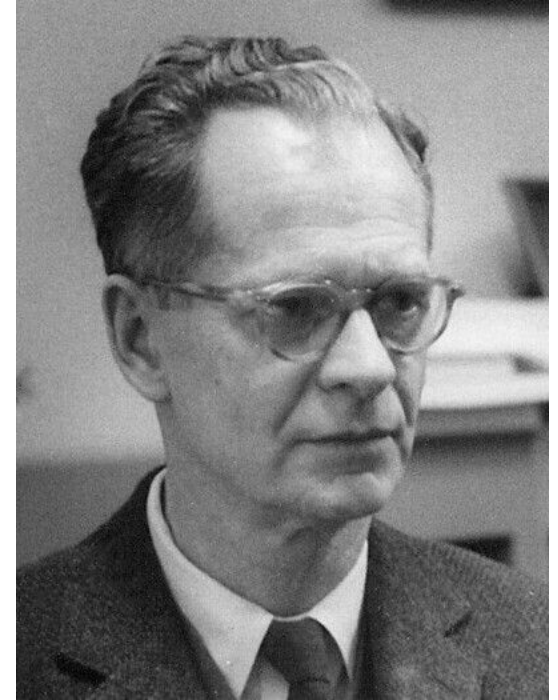
- If the child neglects their homework, the parent might impose an additional chore as a consequence.
- Because this punishment is undesirable, it is less likely the child will avoid homework again.

Skinner: Operant Conditioning



Peck key → Food → More pecking

Behavior → Consequence → Future behavior changes



A Stranger Experiment: “Superstitious” Pigeons

Skinner put hungry pigeons in a chamber and delivered food automatically at regular intervals, regardless of what the pigeon was doing.

What do you think happened?

- One pigeon repeatedly turned counter-clockwise.
- Another thrust its head toward a corner.
- Others developed repeated head or body movements.
- The behavior that happened to occur near food delivery became more likely to recur.



What do you think happened when the feeding time became variable?

- A pigeon has learned that pecking a key can produce food.
- Then later, the timing becomes unpredictable (food only released sometime when pecking happens).

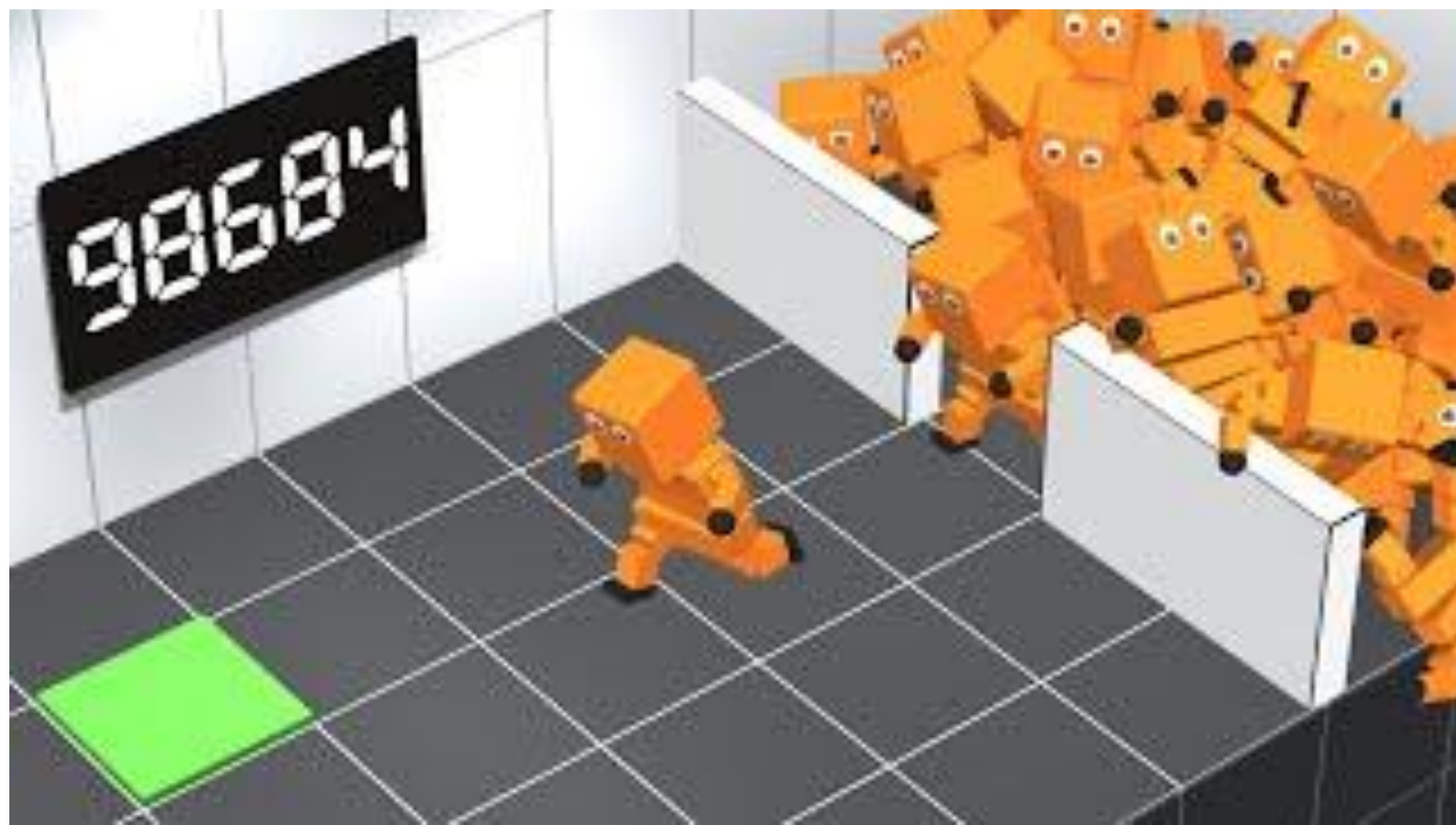
Did pigeon pecking become more or less frequent?

- The pigeon keeps pecking because it cannot predict when the next reward will become available
- What is the human version of repeatedly checking for an unpredictable reward?



AI Connection: Reinforcement Learning

- Reinforcement learning also learns from consequences: actions that produce higher rewards become more likely.
- Like behaviorism, RL focuses on observable behavior and feedback, not necessarily on understanding or subjective experience.
- A system can learn an effective strategy without understanding why it works.
- If rewards are poorly designed, an AI can learn unintended or misleading behaviors – a modern version of “getting the reward signal wrong.”



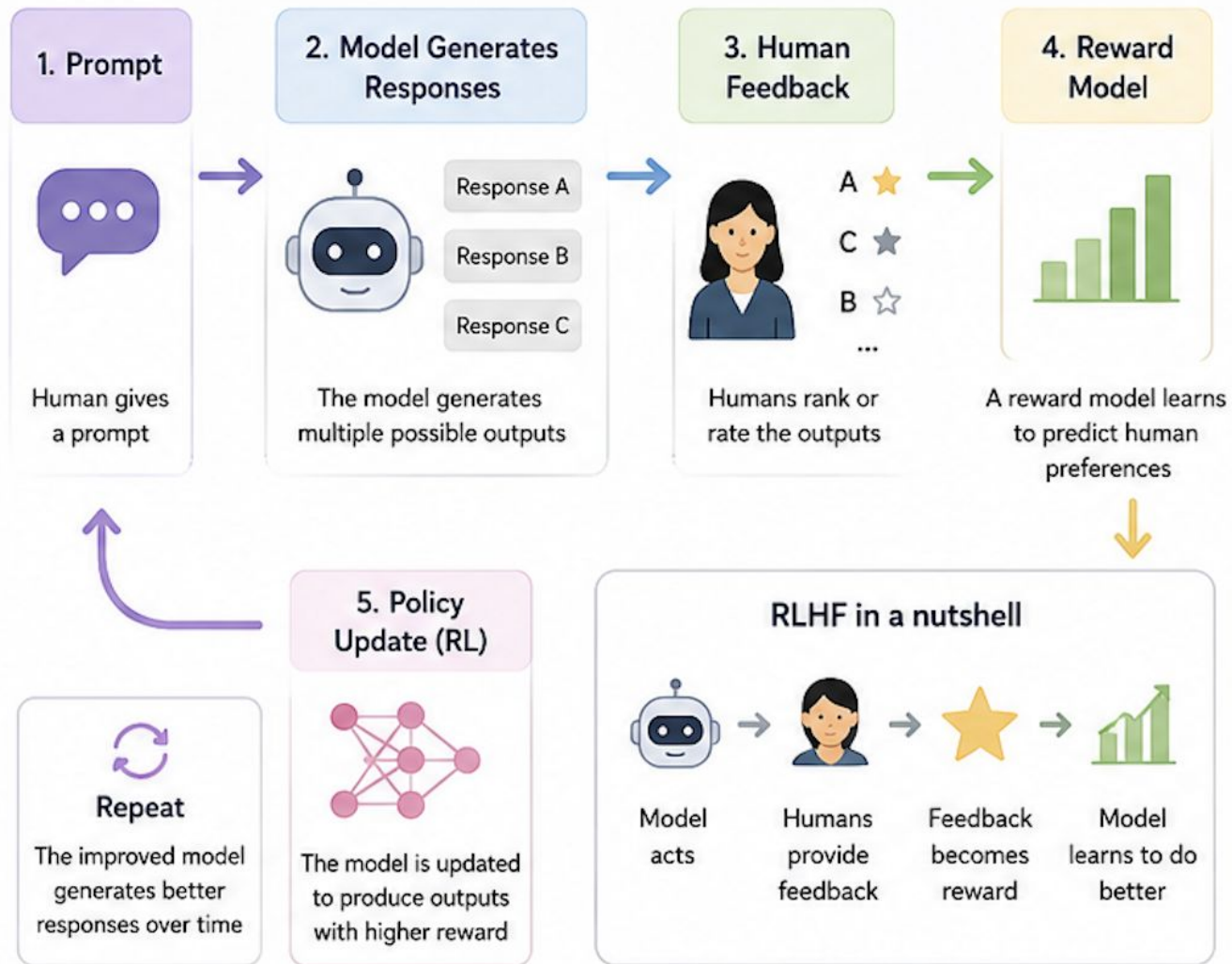
Things to look for...

- **Agent:** Albert: the simulated body being controlled.
- **Environment:** the physical simulation, including gravity, the ground, balance, and movement constraints.
- **Actions:** the motor commands controlling joints/limbs.
- **Reward:** some numerical signal for desirable behavior – especially making forward progress and presumably avoiding falling or inefficient movement.
- **Policy:** the learned strategy that maps Albert's current state to the next action.

The important thing to emphasize is that nobody teaches Albert what walking should look like. The system is not given a sequence like “lift left leg, shift weight, move right leg.” Instead, it tries actions, receives reward, and gradually discovers behaviors that earn more reward.

Example 2: Chatbots + RLHF

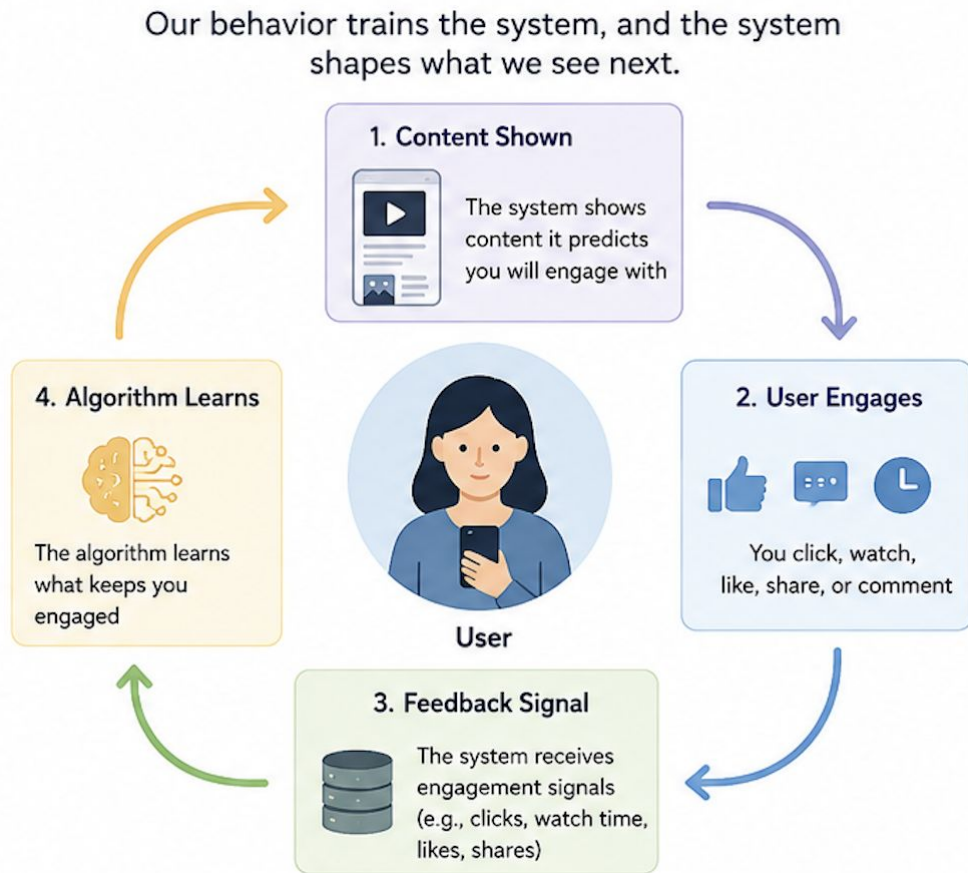
The model learns to produce responses that humans tend to prefer.



Example 3: Silos & Echo Chambers

You see more of the same.
Different perspectives get filtered out. Therefore, the silo / echo chamber gets stronger.

Feedback loops can reinforce our existing behaviors and beliefs, shaping both what we see and what we become.



Behaviorism

Key Concepts:

- *Stimulus-Response*: Learning as a result of responses to stimuli.
- *Reinforcement & Punishment*: Use of rewards and consequences to shape behavior.
- *Trial & error learning*: We learn based on the feedback we receive through our actions.

Notable Theorists:

- B.F. Skinner, John Watson, Edward Thorndike

Discuss: Strengths & weaknesses of reinforcement learning in human beings

Discuss in your groups:

1. What have you learned through repetition, feedback, rewards, or consequences?
2. What kinds of learning seem to require something more than reinforcement?
3. Is changed behavior the same thing as learning? Why or why not?

Discuss: Strengths & weaknesses of reinforcement learning in human beings

- 1.

Behaviorism Critiques

- ***Neglect of “nature”*** – Doesn’t acknowledge our “pre-programmed” cognitive infrastructure (developed through evolution). We’re not blank slates!
- ***Oversimplification of learning*** – can’t explain creativity or language acquisition simply by punishment / reward.
- ***Ethical issues*** – Relying heavily on reinforcement and punishment to control behavior can be seen as manipulative or coercive; can disregard individual autonomy and intrinsic motivations.
- ***Impact on schooling*** – has led to reductive teaching methods, over-emphasis on drill & practice (versus higher-order thinking)

5 minute break!

Activity 2: Figure out the rule

Which belong to the category?

✓ 2, 4, 8

✓ 3, 6, 12

✗ 2, 3, 5

✗ 4, 7, 9

Do these patterns belong in the category? Why?

✓ 2, 4, 8

✓ 3, 6, 12

✗ 2, 3, 5

✗ 4, 7, 9

5, 10, 15

1, 2, 3

1, 2, 4

6, 12, 24, 48,

0, 0, 0

-5, 10, -20

Don't just tell me your answer. What is your current theory of the rule? What evidence would make you change it?

What were you doing to figure out whether the numbers fit?

You weren't simply producing behaviors that I reinforced. You were constructing an explanation, testing it against new experiences, and revising it. That is the theory of **Constructivism**.

Constructivism

Individuals construct knowledge through their experiences and interactions with the world. Learning is seen as an active, contextualized process of constructing knowledge rather than acquiring it.

- **Active Learning** (versus passively receiving information)
- **Prior Knowledge:** Learners' prior knowledge influences their understanding and interpretation of new information (people aren't "empty vessels").

Constructivism: Piaget: Studied his own children

Proposed that learners **actively** construct their understanding of the world through stages of cognitive development.

- Universal stages of cognitive development that all children go through
- Idea of “transfer” – one important goal of learning is to help students develop cognitive resources that generalize to new situations.
 - “Abstract v. Concrete thinking”





Another Big Idea: Assimilation & Accommodation

Assimilation – “This new thing fits into a category I already have in my head.”

Accommodation – “My categories aren’t working anymore. I need to change the categories.”

Going further (next class) – Humans can also question the goals, values, and framing of the problem itself.

AI analogue for Constructivism: Most AI systems are very good at adjusting within a framework that humans already gave them. They are much less able to rethink the framework itself.

Discuss: Changing the answer v. changing the framing

Discuss in your groups:

- How do you know when your existing categories are no longer good enough?
- What is the difference between changing an answer and changing the way you understand the problem?
- Can AI currently “change the frame,” or does it mostly learn within frames humans give it?

Takeaways

Humans may change their categories, their goals, and even question the problem itself.

Modern AIs usually optimizes **within fixed categories, goals, and problems specified during design/training**. How entities get sorted into different categories It might update enormously as a system gets more data. But it usually does not spontaneously decide:

- “Maybe this prediction task isn’t useful.”
- “These categories aren’t the right ones”
- “Maybe I should stop optimizing this objective entirely.”

Example: AI Hiring Tool

Imagine an AI hiring system whose task is:

“Predict which applicants will be successful employees.”

It might update enormously as it gets more data. But it usually does not spontaneously decide:

- “Wait. Maybe ‘successful employee’ is the wrong category.”
- “Maybe this prediction task is socially harmful.”
- “Maybe past hiring outcomes are a bad definition of success.”
- “Maybe I should stop optimizing this objective entirely.”

For Next Time: Constructivist Critiques

- Doesn't take into account cultural differences
- Doesn't consider context
- “Concrete v. Abstract” dichotomy is elitist and invalid

Career Planning: Next Time

- 30% of graduating seniors at UNC Asheville graduate without having spent time planning for what they want to do next, and without having searched for a job or graduate school.
- But it's a lot easier to do these things while still in school because you have a lot of resources still at your disposal.
- You're going to start by looking at yourself, your interests, your values, your aspirations. And then gradually start thinking about how that intersects with various career paths (and how you can learn more).
- Focus is not on technical skills (resumes, cover letters, etc.), but on who you are, what you've gained from their time at UNC Asheville, and how you can share who you are and what you can do more powerfully.